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Ecole polytechnique fédérale de Zurich
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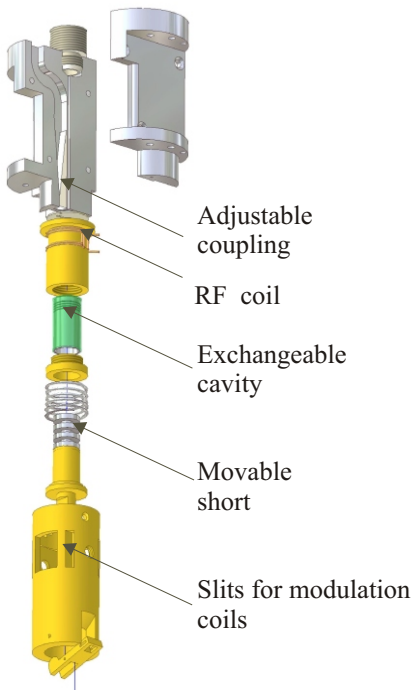
Physical Chemistry
8093 Zürich, Switzerland
http://www.esr.ethz.ch

EPR@ETH: Instrumentation and Methodology

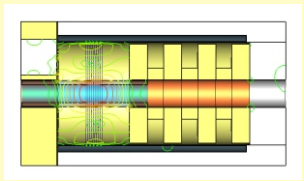
I. Gromov, A. Fedin and A. Schweiger

Probe head at 35 GHz

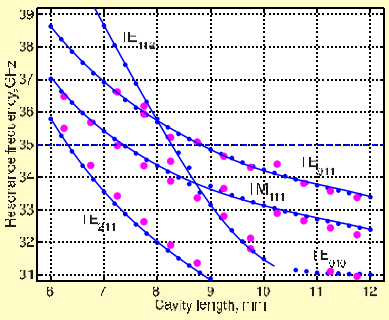
A. Design



B. Simulation



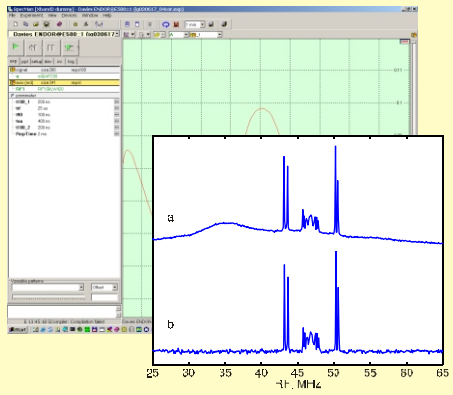
C. Testing



D. Performance

MW conversion factor, Gauss/$W^{1/2}$	2.5
RF conversion factor, Gauss/$W^{1/2}$	0.3
Sensitivity, spins/G/$Hz^{1/2}$	$3x10^9$
Tuning range, GHz	34-36
Sample access, mm	3.9
Q-factor	10^2-10^3

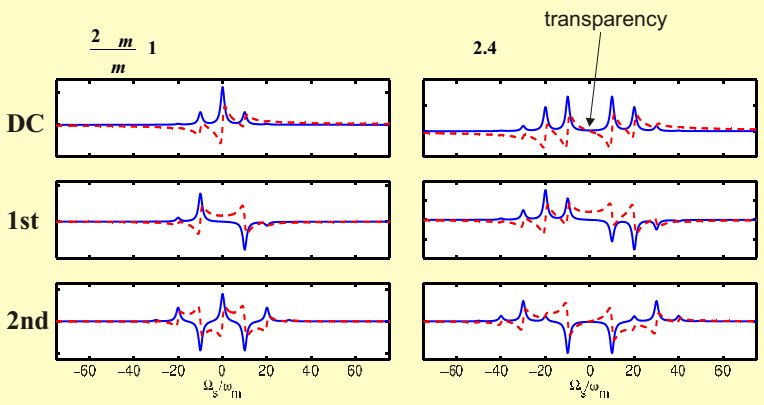
E. Application



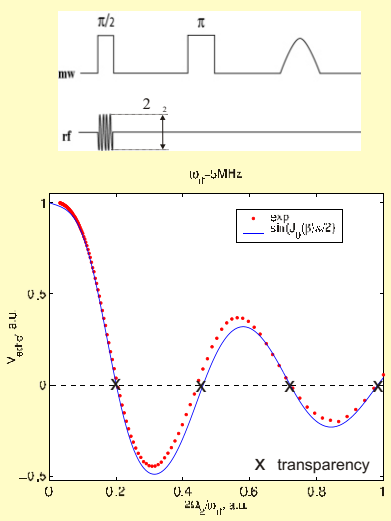
The probe head is based on a cylindrical cavity coupled on top to a waveguide via a Gordon coupling structure (Fig.A). The TE_{011} cavity mode is used for the B_1 B_0 measurements. Following to experimental needs the cavity can be a metallic cylinder or a quartz tube plated with Au/Ag on its inner surface (id 11 mm). In case of ENDOR in the middle of the resonating volume the metallic layer is circularly scratched to make it transparent to the rf field produced by a four-turn saddle coil which is wrapped around the cavity. The influence of the coupling structure, the entry holes and the sample tube size on the cavity eigenmodes and the emf distribution were simulated numerically using MW Studio (CST). The isolines in Fig.B show the magnetic field (TE_{011}) inside the cavity with an 2.9 mm empty quartz tube at 35 GHz. The eigenfrequencies for this case are shown in Fig.C as a function of cavity length. Several modes are observed whose measured frequencies (magenta) coincide well with the simulated ones (blue). The cavity radius and coupling structure is optimized to have only the TE_{011} mode in the working range. As an example of application the proton ENDOR spectra of a Cu-doped glycine single crystal at 15 K are shown in Fig. E. Because relatively low RF conversion factor (Fig. D) 400 W of rf power has been used. Consequently the base line is heavily distorted due to heating effects (a). However applying a random RF sweep using a newly developed spectrometer control program (see www.esr.ethz.ch) the distortions are completely removed (b).

Field-modulated EPR

A. CW case



B. Pulse EPR



C. Toggling frame

$$H_{lab}(t) \quad s+2 \quad m\cos(\quad mt)\ S_z + 2 \quad 1\cos(\quad mwt)S_x \qquad (1)$$

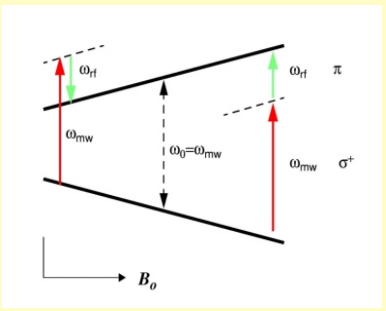
$$\Downarrow$$

$$R(t) \quad e^{\,i\ k\ mt\ \frac{2\ m}{m}\sin(\quad mt)\quad mw\ S_z} \qquad (2)$$

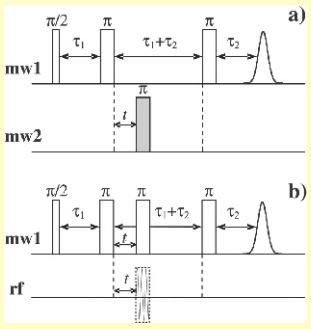
$$\Downarrow$$

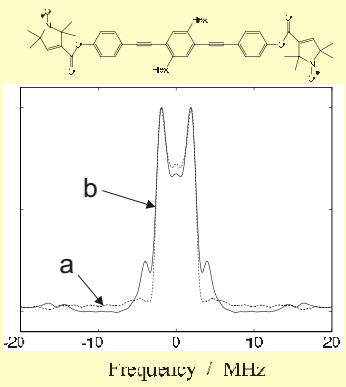
$$H_{tog,k} = (\quad s - k\quad m)S_z + J_k\ \frac{2\ m}{m}\quad 1S_x \quad O(t) \qquad (3)$$

It is well known that cw field-modulated EPR spectra can have artificial structures when the modulation frequency $_m$ exceeds the width of the resonance line as is shown on left panel in Fig. A. These spectra are simulated using a very small modulation index $^2\ m/_m \quad 1$, where $2\ _m$ is the amplitude of the modulation field. Several sidebands are observed for DC and at the first and second harmonic of the signal. The harmonic intensities are weighted with corresponding first kind Bessel functions. In case of the DC contribution the factors are simply J_n^2 . The Bessel functions have zero-crossings so that different sidebands disappear at certain value of e.g. the central band ($_s$=0) reduces to zero at (Fig. A right). The origin of the sidebands becomes clear when one uses Floquet theory to study the evolution of the spin system under a periodical spin Hamiltonian (1) or by using the second quantization approach [1, 2]. The sidebands are due to multiple-photon transitions $^+ + k$, where one mw $^+$photon and k photons of the modulation field are simultaneously absorbed or emitted. These pecularities of spin behavior led us to the idea to use longitudinally oscillating fields in pulse EPR [1, 3]. For example to control the spin excitation by the tuning frequency or amplitude of the dressing field. Fig. B shows how the primary echo amplitude depends on the strength of the rf field applied simultaneously with the /2 mw pulse. The echo amplitude (circles) follows a slightly dumped $\mathbf{sin}(J_0(\quad)\ /2)$ curve (solid line) in accordance with the Hamiltonian (3) given in C. At zero-crossings the system becomes transparent to the mw field. Recently, this strategy was applied to the DEER experiment, which requires two mw frequencies as is shown in Fig. D (a) [4]. The pulse on the second mw frequency has been replaced by a composite mw/rf pulse (b). The rf amplitude was tuned to have a small effective field on the transition resonant with mw1 and maximal field on the *first sidebans* of this transition. The comparison of two experiment approaches (a) and (b) shows that the new sequence effectivelly substitute the old one.



D. Application: New DEER sequence





[1] I. Gromov and A. Schweiger. JMR **146,** 110 (2000)
[2] M. Kälin, I. Gromov and A. Schweiger. JMR **160**(2), 166 (2003)
[3] M. Kälin, I. Gromov and A.Scweiger. PRA **69**, 033809 (2004)
[4] M. Fedin, M. Kälin, I. Gromov and A. Schweiger. JCP **120**(3), 1361 (2004)